

Review Article

Disentangling research on oak decline factors in Mediterranean-type climate regions: A systematic review

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ABSTRACT

The genus *Quercus* holds significant ecological and economic value in the Northern Hemisphere Mediterranean-type climate (MTC) regions. However, *Quercus* species and ecosystems are threatened by decline. Despite the importance of this genus, a comprehensive analysis of the causes of oak decline across these regions is still lacking. This study maps and analyzes research on oak decline in MTC regions, examining causal factors and *Quercus* species studied, while identifying knowledge gaps and future research priorities. We systematically reviewed 241 peer-reviewed articles on oak decline in MTC regions up to 2022, retrieved from the Web of Science and Scopus databases.

Our findings show that scientific articles on oak decline began to be published in 1981 and peaked in 2021. Most studies focused on the Mediterranean Basin, with disparities between Northern Africa and Southern Europe. The most frequently studied species were *Quercus suber* L. and *Quercus ilex* L. in the Mediterranean Basin, and *Quercus agrifolia* Née and *Quercus kelloggii* Newb. in California, comprising 53% of all species mentioned. Pathogens (48%) and climate factors (17%) were identified as the most researched causes of oak decline. Additionally, multiple interactions between factors were identified, confirming that these combinations potentially exacerbate oak decline. Climate factors were the most frequently found in combination with others. Conversely, pathogens were mostly studied in isolation, reflecting their widely recognized role in oak decline.

The study highlights the potential risk of pathogen and pest transference between MTC regions and the critical role of management practices in influencing oak decline in interaction with biotic and abiotic factors.

1. Introduction

The genus *Quercus* is one of the most widespread and diverse among trees in the Northern Hemisphere, spanning from the equator to the boreal regions (Kremer and Hipp, 2020). It comprises 435 different species (Denk et al., 2017), which includes dominant overstory trees as well as shrubs and small trees. The diversity, economic importance (Nixon, 2006) and environmental value (Scarascia-Mugnozza et al., 2000) of *Quercus* species are particularly significant in the two Mediterranean-type climate (MTC) regions of the Northern Hemisphere: the Mediterranean Basin and California (which includes parts of southwestern USA-California, and northwestern Mexico-Baja California, here designated as California, for simplicity) (Denk et al., 2017; Gil-Pelegrín et al., 2017b). Despite the existence of five regions in the world with Mediterranean-type climate (MTC) (Archibold, 1995; Esler et al., 2018), oak species only occur in the two MTC regions of the Northern Hemisphere. These two regions (hereafter MTC regions) are

characterized by hot and dry summers, cool and moist winters, and a distinctive summer drought (Archibold, 1995; Esler et al., 2018; Siegmars-Walter Breckle, 2002). Although annual precipitation and length of the dry season may vary widely within these regions, their vegetation and biotopes have experienced co-evolution and exhibit several ecological similarities (Archibold, 1995; Siegmars-Walter Breckle, 2002).

The Mediterranean Basin has approximately 40 oak species (Blondel and Aronson, 1999), while California hosts around 25 oak species (Nixon, 2002; Valencia-A., 2004). Despite geographical distances, oak-dominated landscapes in California and the Mediterranean Basin share similarities in geography and climate, management, and transformative human use over millennia (Alagona et al., 2013; Anderson and Keeley, 2018; Bugalho et al., 2011). Different oak species have developed adaptations to the Mediterranean climate, such as evergreen sclerophyllous leaves, reduction of leaf size, and the development of leaf waxes and trichomes (Gil-Pelegrín et al., 2017b).

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Despite their importance, oak species are threatened by decline across MTC regions. There are references to oak decline in the Mediterranean Basin as early as 1898 (Baeta Neves, 1954; FAO and Plan Bleu, 2018; Vallejo et al., 2009) and for California since the 1990s (Huesca et al., 2021; Rizzo and Garbelotto, 2003). Tree decline refers to a progressive and visible loss of tree vigor and health, which involves a series of observable trait alterations, often resulting from a complex interplay of biotic and abiotic factors (Ciesla and Donaubauer, 1994). Although it frequently culminates in tree mortality (Manion, 1981), definitions of decline vary (Ciesla and Donaubauer, 1994; Manion, 1981; Rodríguez-Calcerrada et al., 2017; Thomas et al., 2002). In some regions, tree death may appear to occur rapidly or suddenly; in other regions, the progression from initial decaying symptoms to mortality can span from months to years (Rizzo and Garbelotto, 2003). In any case, oak decline leads to a reduction in biomass, productivity, or benefits from the forest, contributing to forest degradation (IPBES, 2018). For this study, we adopted a broad definition of decline, encompassing both individual tree issues to forest-wide changes in composition and structure, while also considering the impacts of land-use change.

In MTC regions, oak decline has been attributed to climatic factors (drought and extreme temperatures) (Allen et al., 2010), pathogens (Camilo-Alves et al., 2013; Rizzo and Garbelotto, 2003), pests (Tiberi et al., 2016), or management practices (Bugalho et al., 2011). However, to our knowledge, no comprehensive review has been conducted to synthesize oak decline drivers across MTC regions.

A systematic review is a literature review that aims to summarize the evidence on a topic, in a transparent, repeatable way (Haddaway and Pullin, 2014) and is a recognized standard for synthesizing information on health and environmental themes (Collaboration for Environmental Evidence, 2022). In the present study, we use a systematic review aiming to: (1) map scientific research on oak decline across the MTC regions; (2) provide an overview of the main causal factors of oak decline and the main *Quercus* species studied in MTC regions; (3) analyze the geographical distribution of identified oak decline factors within MTC regions; (4) identify common and different oak decline factors within MTC regions and among countries, as well as their interactions; (5) identify knowledge gaps and future research needs contributing to a better understanding of oak decline and to suggest prevention and mitigation measures.

The lack of comprehensive synthesis of oak decline drivers across Mediterranean-type climate (MTC) regions in the Northern Hemisphere hinders the development of effective management strategies and policy decisions aiming at mitigating this phenomenon. By conducting a systematic review of scientific literature on the topic of oak decline in these regions, this study seeks to provide insights for policymakers, forest managers, and ecologists to better understand and preserve oak forests globally and to formulate informed decisions.

2. Materials and methods

2.1. Search strategy

This review focuses on the Mediterranean Basin and California, which are the MTC regions where the genus *Quercus* is present. Preliminary search string tests in Web of Science (WoS) and Scopus revealed that the term “Mediterranean” was not a keyword in some relevant research papers, resulting in their exclusion from an initial search. Consequently, we expanded the search criteria by including the names of all countries surrounding the Mediterranean Sea, excluding Egypt (because it does not have a MTC region) and including Portugal. Additionally, we used the term “California” to include both the United States of America state of California and the Mexico state of Baja California.

After a first search on “oak decline”, we found that “decline” was often employed interchangeably or as a synonym for various other words that describe similar phenomena. These words included “mortality”, “dieback”, “death”, “degradation”, and “deforestation”. Hence,

in this review, as previously stated, we defined oak decline as a broad concept that included all the words referred to above to encompass the range of negative impacts experienced by mature oak trees that can lead to the loss of tree or stand vigor and health, or ultimately to tree death and which, consequently, will affect the oak stand.

Considering the above-mentioned regions/countries and the definition of oak decline, we therefore used the following search string:

(oak* OR quercus) AND (mediterranean* OR california OR portugal OR spain OR france OR italy OR slovenia OR croatia OR bosnia and herzegovina OR montenegro OR albania OR greece OR turkey OR syria OR cyprus OR lebanon OR israel OR tunisia OR algeria OR morocco OR libya) AND (“decline” OR “mortality” OR “dieback” OR “death” OR “degradation” OR “deforestation”)

The search was performed on the 8th of May 2023 in WoS and Scopus, by searching for the selected words in the article’s title, abstract, or keywords, with no temporal limitation. The WoS search returned 2222 results, while Scopus yielded 1701. Subsequently, all the 3923 references were downloaded and checked for duplicates using ZOTERO (Roy Rosenzweig Center for History and New Media, 2023). Automated duplicate identification by ZOTERO, followed by a manual check, resulted in the removal of 1330 duplicated papers. Further refinement excluded 116 additional references that were not published in peer-reviewed journals, such as conference papers, reports, books, and book sections. We analyzed peer-reviewed articles published in scientific journals, written in English, and excluding review papers. This resulted in 2477 references being selected for a first screening evaluating the title and abstract.

2.2. Screening process: inclusion and exclusion criteria

The screening process followed predefined inclusion and exclusion criteria. First, studies were selected if they examined *Quercus* species located within MTC regions in the countries previously identified. We then selected papers identifying causal factors contributing to the oak decline, at either the individual tree level or at the stand level. Conversely, studies were excluded from the analysis if: (i) focused solely on seeds, seedlings, and saplings, unless demonstrating effects that could be extrapolated to adult oak trees; this exclusion was made as oak regeneration and seedling recruitment have already been reviewed elsewhere (Pulido, 2002; Tyler et al., 2006); (ii) relied exclusively on species distribution models or predictive modeling approaches without incorporating original data collection.

Because of this initial screening, we selected 403 papers for a final screening involving a thorough examination of the full text and the application of the eligibility criteria previously described. This final screening resulted in a total of 241 papers selected for data extraction. We considered only articles with a publication date up to and including 2022, to cover all studies made publicly available on an annual basis. The search and screening procedure are shown in (Fig. 1) (adapted from Haddaway et al., 2017).

2.3. Data extraction and textual network analysis

Following the selection process, we retrieved information from the resulting articles, namely: bibliographic information, date of decline detection, study location, climatic and physical characteristics of the study location, study design (experimental or observational), species studied (oaks and others), and causal factors of decline (Table 1).

We first identified and collected all individual causal factors of decline from the selected articles. We used the term “occurrence” to denote when a specific causal factor was mentioned in a given study since more than one factor could be identified in one article.

Then, to assess the research focus on different aspects of oak decline, we classified the causal factors collected in each selected article using the set of categories previously shown in Table 1, which encompassed various decline factors (Table 2). We counted each category only once

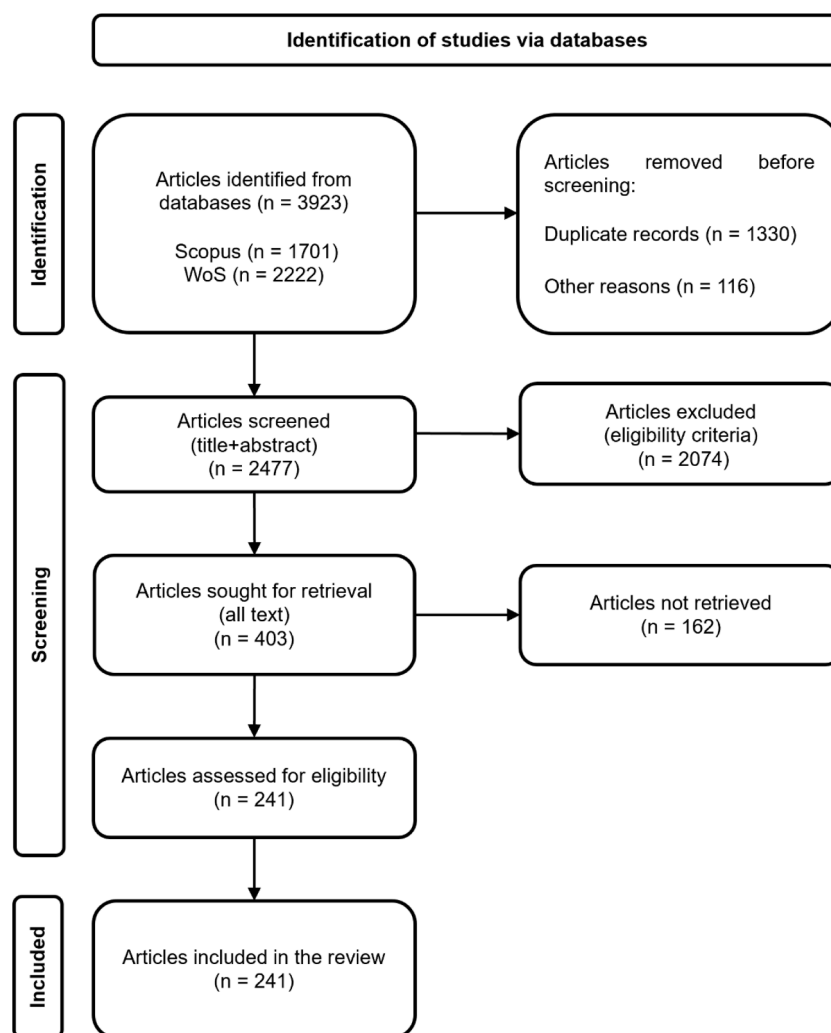


Fig. 1. Procedure of literature search and screening for the systematic review.

per study, regardless of how many specific factors within that category were investigated, to avoid overrepresentation. For example, if a study examined multiple pathogens, the 'pathogen' category was counted only once for that study. This approach allowed us to represent the overall research effort dedicated to each category of decline factors across the literature, providing insight into which aspects of oak decline have received the most attention in research. The database with all the articles selected for this review is available as a spreadsheet in Supplementary Material Data 1.

Lastly, we used a textual network analysis (Segev, 2020) to build a co-occurrence network between the factor categories that were identified in the selected articles. Textual network analysis is a method for content analysis that focuses on the interactions between nodes (categories in this case), to identify, quantify, and understand patterns of data (Segev, 2020). We used this method to identify the interactions between the categories analyzed in each article by quantifying the frequency of co-occurrences, i.e., the frequency in which different categories appear together in each article. We used the *igraph* package (Csardi and Nepusz, 2006) for R software (R Core Team, 2021) to build the co-occurrence network.

3. Results

3.1. Temporal trend of research and onset of oak decline in MTC regions

Our analysis revealed that the first research article on oak decline in

MTC regions was published in 1981, with a substantial increase in research after 2008, mainly driven by studies conducted in the Mediterranean Basin (Fig. 2). The highest number of articles was published in more recent years, namely in 2017 and 2021, with 21 and 24 publications, respectively. Notably, no article was selected regarding the Baja California region (Mexico), therefore, hereafter California refers to the region covered by the state of California in the USA.

We found information regarding the date of onset of oak decline in 138 papers (Fig. 3). More than half of these papers (54%, $n = 75$) identified the 1980s and 1990s as the initial period of oak decline observation, followed by the first two decades of the 21st century (29%, $n = 40$). We found a notable difference in the timing of initial decline observations between MTC regions: in the Mediterranean Basin, the 1980s were cited as the main starting point for oak decline; in contrast, the articles for California showed the onset of oak decline predominantly in the 1990s and subsequent decades.

3.2. Geographic focus

The studies selected by this review comprised research located in 12 countries (Fig. 4) out of the 23 initially stated. The majority of the selected articles (81%) correspond to study locations within the Mediterranean Basin region, particularly in Spain (32%, $n = 82$), Italy (18%, $n = 45$), and Portugal (16%, $n = 41$). The United States of America emerges as the second most studied country, with 49 (19%) articles. Concerning Northern Africa, the highest number of papers were focused

Table 1
Information retrieved from the selected articles.

Category	Description	
Bibliographic information	Data related to the publication details of the article (DOI, Authors, Year published; Title; Journal; Volume; Issue; Pages)	
Date of decline detection	Date when oak decline was initially observed/reported in the study area by the author or recognized in supporting literature cited	
Study location	Geographical location of the study (country and region)	
Climatic and physical characteristics of the study location	Data related to climate, soil, and topographic conditions of the study area	
Study design and data collection	Data related to the study design and data collection methods	
Species studied	Data related to the <i>Quercus</i> species studied, including their names and leaf habits (evergreen or deciduous) and identification of other tree species included in the study (not <i>Quercus</i>)	
Causal factor of oak decline:		
Factor(s) causing decline identified in the study	<i>Factor identification</i>	Name of the decline factor identified
	<i>Factor category</i>	Classifies the decline factor based on its characteristics or origin into nine categories: climate, fire, human activity, pathogens, pests, soil, topography, tree/stand characteristics, and “others”
	<i>Factor type</i>	Nature of the decline factor distinguishing between biotic and abiotic
	<i>Number of factor categories</i>	Number of factor categories identified: one agent or multi-agent

Table 2
Classification and identification of oak decline factors retrieved from selected articles.

Factor category	Definition of decline factors included
Climate	Factors related to elevated temperatures, drought events, precipitation patterns (including heavy rains concentrated in short periods and waterlogging)
Fire	Any occurrence or impact of fire events
Human activity	Anthropogenic factors including forest management practices, land-use changes, and various forms of pollution
Pathogens	Disease-causing organisms (e.g. fungi, Oomycota, bacteria, viruses)
Pests	Insects belonging to the class Insecta
Soil	Factors related to soil properties and soil types
Topography	Landscape features including aspect, slope, and elevation
Tree/stand characteristics	Factors encompassing morphological or physiological traits (e.g., bark thickness, tree size) and stand characteristics (e.g., pure vs mixed species, understory composition)
Others	Any additional factors not fitting into the above categories

on Tunisia, with 10 papers, followed by Algeria and Morocco with four papers published on each country. There were seven articles that included study locations in more than one country in the Mediterranean Basin.

3.3. *Quercus* species studied

We identified 36 oak species in the selected studies (Table 1 of Supplementary Data 2). *Quercus suber* L. was the species most addressed (21%, $n = 98$ articles) followed by *Quercus ilex* L. (19%, $n = 88$ articles), both evergreen species native to the Mediterranean Basin. Species native to the California region, namely *Quercus agrifolia* Née (8%, $n = 38$ articles) and *Quercus kelloggii* Newb. (5%, $n = 23$ articles) ranked third and sixth, respectively. Although we found a greater diversity of deciduous

oak species in the selected articles (28 out of 36 species), the most studied *Quercus* species were evergreen (57% of total species), as stated above.

3.4. Causal factors of oak decline

We found 205 distinct causal factors contributing to oak decline, related to different categories such as pathogens, pests, climate, or management, with the highest number reported for pathogens (282 occurrences – 143 species/factors identified), followed by climate (106 occurrences – 3 different factors identified) (Table 3). Recall that we used the term “occurrence” to denote every specific causal factor that was mentioned in a given study since more than one factor could be identified in one article. Full details of causal factors identification are available in Supplementary Material Data 1.

Among pathogens, fungi and Oomycota species emerged as the most frequently investigated, representing 26% ($n = 198$) and 19% ($n = 137$) of total occurrences, respectively. Notably, we found no reference to viruses causing oak decline. Fungi exhibited the highest diversity among pathogens, with 40 different families investigated. Within the fungi, the highest number of articles addressed the *Botryosphaeraceae* family, with 74 occurrences and 25 different species. Three of the investigated species (*Botryosphaeria corticola* (anamorph: *Diplodia corticola*), *Botryosphaeria sarmentorum* (anamorph: *Dothiorella sarmentorum*), and *Dothiorella iberica*) are common to the Mediterranean Basin and California. We also found articles mentioning *Fusarium solani* (*Nectriaceae*), *Armillaria gallica* and *Armillaria mellea* (*Physalacriaceae*) in the two regions. In the Mediterranean Basin, *Botryosphaeria corticola* (anamorph: *Diplodia corticola*) and *Biscogniauxia mediterranea* were the most investigated fungi in the selected articles. Oomycota, though less diverse than fungi, were represented by two families: *Peronosporaceae* and *Pythiaceae*. Within *Peronosporaceae*, *Phytophthora* was the most studied genus globally (18%, $n = 130$). Regional differences were evident in this genus: in California, *Phytophthora ramorum*, the causal factor of “sudden oak death,” dominated the selected articles for this region (20%, $n = 26$), with *Phytophthora nemorosa* and *Phytophthora pseudosyringae* being addressed in only two articles. In contrast, in the Mediterranean Basin, *Phytophthora cinnamomi* was the most frequently investigated (7%, $n = 39$). This region was also associated with a higher diversity of studied *Phytophthora* species (27 species).

Climate factors were the second most addressed causal factor of oak decline (17%; $n = 124$). Within this category, drought was the most frequently researched factor (61%, $n = 76$). Pests ranked third, with 12% ($n = 85$) of total occurrences, encompassing 11 families. The *Curculionidae* was the most frequently referred family, with 35 occurrences, followed by *Buprestidae* (19) and *Cerambycidae* (13). Regional differences were observed with distinct species in the Mediterranean Basin and California. Specifically, the *Cerambycidae* family was only studied in the Mediterranean Basin and there were no common species of the *Buprestidae* family investigated in both regions. Only one pest species was addressed in both regions: the ambrosia-beetle *Xyleborinus saxesenii*. Human activity factors, which included forest management practices, land-use changes, and various forms of pollution, were the fourth most referred category of causal factors (7%, $n = 48$), with management practices being the most frequently addressed ($n = 33$). Topography factors were studied in 6% ($n = 40$) of total occurrences, especially slope ($n = 21$) and aspect (exposure) ($n = 15$). Other factors, including tree/stand characteristics, fire, soil, and miscellaneous decline-related factors (included in the category “others”), comprised the factors investigated in the remaining articles.

3.5. Research focus per category of oak decline

After identifying the diverse individual causal factors identified in selected articles, we investigated the frequency of the categories present in those studies. This analysis allowed us to evaluate the research focus

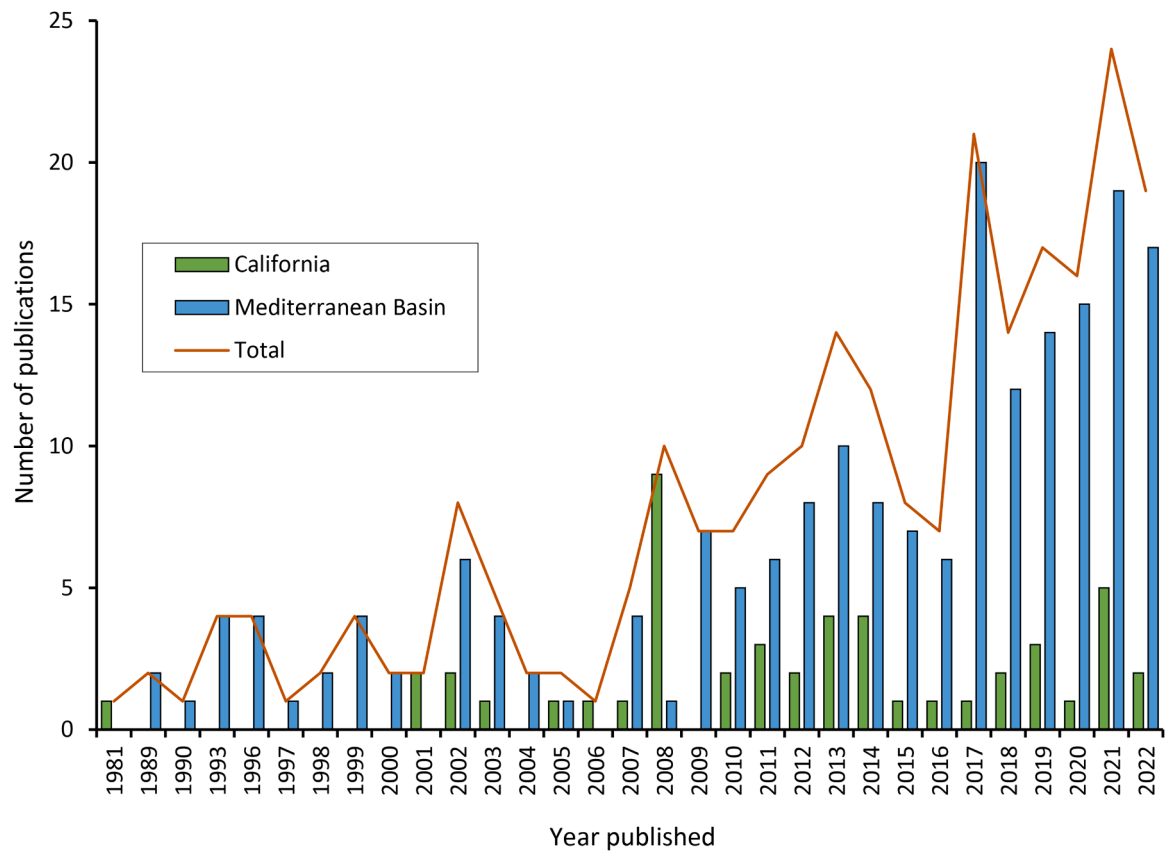


Fig. 2. Number of selected publications in MTC regions in the period 1981–2022.

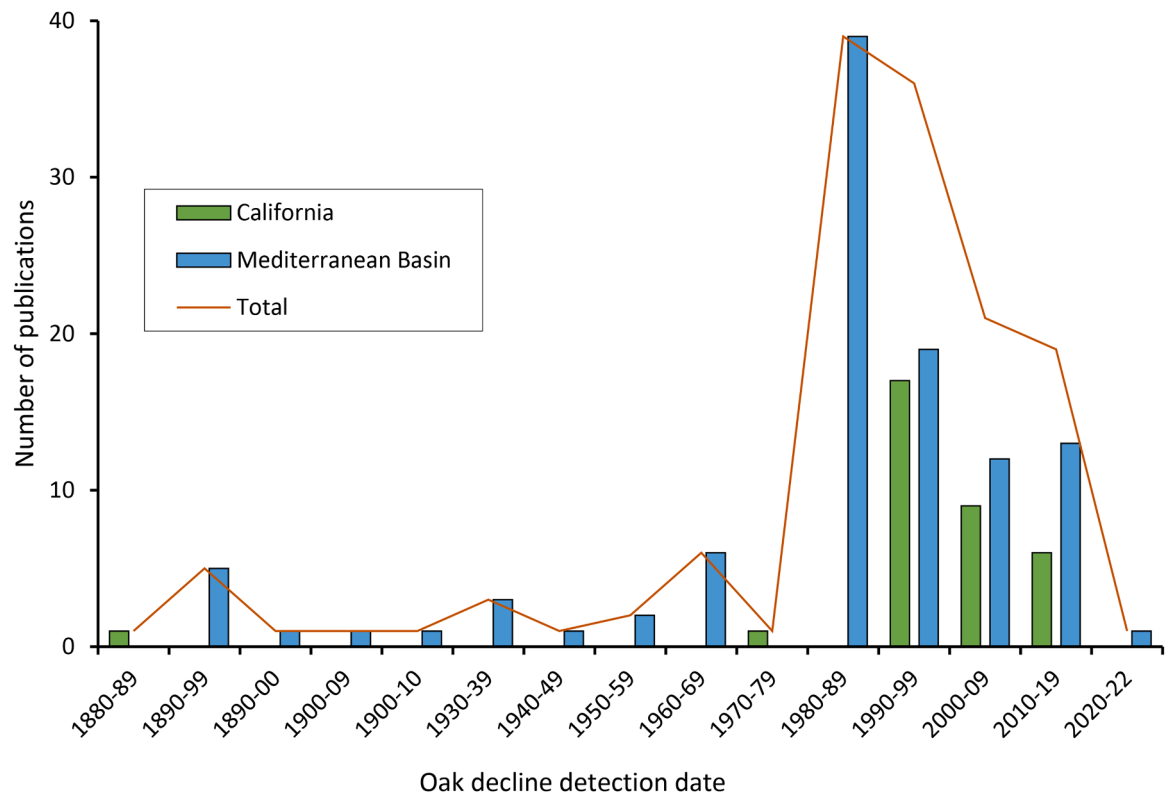


Fig. 3. Number of selected publications that mention the date of onset of oak decline detection.

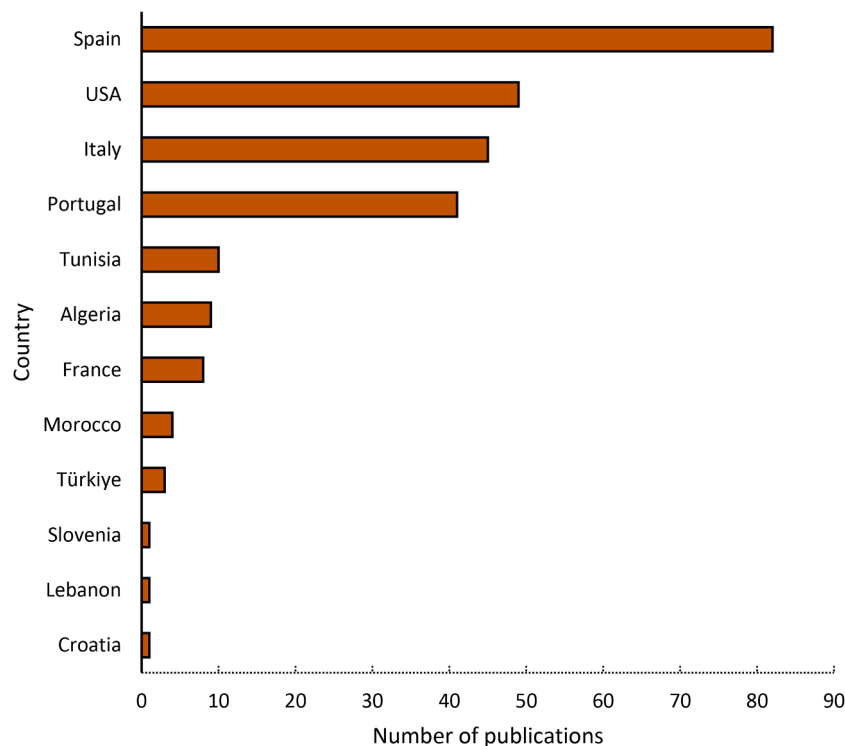


Fig. 4. Geographic distribution of selected articles on oak decline in MTC regions.

on different aspects of oak decline. The figure below (Fig. 5) shows that pathogens were the most studied factor category in both regions, accounting for 33% ($n = 133$) of the total articles, followed by climate-related factors, which made up 22% ($n = 88$). Interestingly, research efforts were very similar between MTC regions for most of the categories, with the exception of pathogens, which have more dedicated studies in California, and human-activity related factors, which are most prevalent in the Mediterranean Basin.

A temporal analysis of the distribution of the factor categories (Fig. 1, Supplementary Data 2) revealed a diversification in the categories analyzed since the beginning of research on this topic, with fewer categories being addressed by papers from the 1980s to 90s, and a higher number of papers addressing seven to eight categories in the 2020s. Approximately half of the selected studies examined multiple categories of factors (49%, $n = 119$), while 51% ($n = 122$) focused on a single category. Biotic factors were the most frequently investigated (49%, $n = 119$), followed by studies addressing both abiotic and biotic factors categories (34%, $n = 81$). Studies focusing solely on abiotic factors comprised 17% ($n = 41$).

Investigated causes of decline varied widely across countries. For example, in Spain, climate was the most frequently studied category followed by pathogens (Fig. 2, Supplementary Data 2). In California, pathogens were the most frequently researched category, followed by pests and climate. In Italy, pathogens were also the most frequently researched category, followed by climate. As for Portugal, research on oak decline focused primarily on the category of human activity, followed by pathogens.

3.6. Network analysis of causal factor categories of oak decline

The textual network analysis of the categories identified in the articles revealed interactions (co-occurrences) between various causal factor categories (Fig. 6). Pathogens were most often investigated in association with climatic factors ($n = 26$) and pests ($n = 17$). This category was also found in co-occurrence with human activity and tree/stand characteristics (both $n = 9$), as well as soil and topography (both n

$= 8$). Climatic factors showed the highest frequency of interactions in a larger number of categories. Besides being investigated together with pathogens ($n = 26$), as seen above, they were frequently analyzed alongside human activity ($n = 16$), soil ($n = 13$), tree/stand characteristics ($n = 13$), topography ($n = 10$) and pests ($n = 7$). Human activity-related factors, beyond their association with climate ($n = 16$), co-occurred with fire and topography (both $n = 12$), and pathogens ($n = 9$). Notably, pests were found to be most frequently studied in conjunction with pathogens.

Pathogens stand out as the most frequently category studied in isolation ($n = 76$ studies), meaning they are often examined without considering interactions with other factors. This is followed by climate ($n = 28$) and human activity ($n = 11$) (see also Table 2 – Supplementary Data 1, for detailed information).

4. Discussion

4.1. Temporal and spatial trends in oak decline research

Our results identified the first published papers on oak decline in the 1980s and 90s, suggesting a reactive response of the scientific community to emerging concerns related to symptoms of degradation of oak systems. This period saw a series of symposia and comprehensive reviews on oak decline (Ciesla and Donaubauer, 1994), coinciding with the identification of *Phytophthora* pathogens as main contributors to decline in Europe (Brasier et al., 1993) and California (McPherson et al., 2001; Rizzo and Garbelotto, 2003).

The geographical distribution of research efforts revealed significant disparities. Most oak decline research has been conducted in the Mediterranean Basin, particularly in Spain, Italy, and Portugal, with a much lower number of studies in Northern African countries. This disparity may reflect differences in research funding between European and African Mediterranean countries (Rivero, 2024).

In California, research has been mostly concentrated in coastal areas, particularly in counties adjacent to San Francisco County, such as Marin County and Monterey County. Notably, we found no studies addressing

Table 3
Number of occurrences of oak decline causal factors in the selected papers.

Oak decline causal factor	Mediterranean basin	California	Total
PATHOGENS – (143 species/factors)	282 (48%)	60 (47%)	342 (48%)
Fungi – 106 species	168 (29%)	30 (23%)	198 (28%)
<i>Botryosphaeriaceae</i> – 24 species	64 (11%)	10 (8%)	74 (10%)
<i>Nectriaceae</i> – 7 species	10 (2%)	2 (2%)	12 (2%)
<i>Graphostromataceae</i> – 1 species	11 (2%)	0 (0%)	11 (2%)
<i>Ophiostomataceae</i> – 8 species	10 (2%)	1 (1%)	11 (2%)
36 other families – 66 species	73 (12%)	17 (13%)	90 (13%)
Chromista (Oomycota) – 31 species	108 (18%)	29 (22%)	137 (19%)
<i>Peronosporaceae</i>	101 (17%)	29 (22%)	130 (18%)
<i>Phytophthora cinnamomi</i>	39 (7%)	0 (0%)	39 (5%)
<i>Phytophthora ramorum</i>	1 (0%)	26 (20%)	27 (4%)
29 other species	61 (10%)	3 (2%)	64 (9%)
<i>Pythiaceae</i> – 3 species	7 (1%)	0 (0%)	7 (1%)
Bacteria – 6 species	6 (1%)	1 (1%)	7 (1%)
CLIMATE – (3 factors)	106 (18%)	18 (14%)	124 (17%)
drought	69 (12%)	7 (5%)	76 (11%)
temperature	32 (5%)	6 (5%)	38 (5%)
precipitation	5 (1%)	5 (4%)	10 (1%)
PESTS (CLASS INSECTA) – (48 species/factors)	58 (10%)	27 (21%)	85 (12%)
<i>Curculionidae</i> – 15 species	22 (4%)	19 (15%)	41 (6%)
<i>Buprestidae</i> – 12 species	14 (2%)	5 (4%)	19 (3%)
<i>Cerambycidae</i> – 11 species	13 (2%)	0 (0%)	13 (2%)
7 other families – 10 species	9 (2%)	3 (2%)	12 (2%)
HUMAN ACTIVITY – (3 factors)	45 (8%)	3 (2%)	48 (7%)
management	32 (5%)	1 (1%)	33 (5%)
land-use/land changes	10 (2%)	2 (2%)	12 (2%)
pollution	3 (1%)	0 (0%)	3 (0%)
TOPOGRAPHY – (3 factors)	32 (5%)	8 (6%)	40 (6%)
slope	16 (3%)	5 (4%)	33 (5%)
aspect	13 (2%)	2 (2%)	12 (2%)
elevation	3 (1%)	1 (1%)	3 (0%)
TREE/STAND STRUCTURE – (2 factors)	23 (4%)	7 (5%)	30 (4%)
tree structure and physiology	19 (3%)	6 (5%)	25 (3%)
stand characteristics	4 (1%)	1 (1%)	5 (1%)
FIRE – (1 factor)	19 (3%)	5 (4%)	24 (3%)
SOIL – (1 factor)	21 (4%)	1 (1%)	22 (3%)
OTHERS – (1 factor)	2 (0%)	0 (0%)	2 (0%)
Grand Total	588 (100%)	129 (100%)	717 (100%)

oak decline in Mexico's Baja California region. These geographical gaps in research could overlook endemic oak species and local ecotypes, which may lead to future ecosystem loss (Scarascia-Mugnozza et al., 2000).

In addition, our findings showed a diversification in the categories analyzed since the beginning of research on this topic. This reveals a different perspective of the scientific community moving from a single to multiple causes of oak decline or a better perception within the research community on more detailed causes of decline within each category.

4.2. Ecological and socio-economic drivers of oak decline research

Our analysis of oak decline research revealed distinct ecological and socio-economic drivers in the Mediterranean Basin and California. These drivers are reflected in the number and types of oak species studied, as well as in the focus on specific species of high ecological and economic importance. We found a higher number of studied oak species in the Mediterranean Basin ($n = 31$) compared to California ($n = 9$). This could be expected due to a higher number of *Quercus* species in the Mediterranean Basin (40) than in California (20). Nevertheless, Mediterranean Basin studies also proportionally addressed a higher number of species. The majority of these studies addressed evergreen species, particularly

Quercus suber L. and *Quercus ilex* L., which dominate research in Portugal and Spain.

The two most studied oak species in each region (*Quercus suber* L., *Quercus ilex* L. in the Mediterranean Basin and *Quercus agrifolia* Née, *Quercus kelloggii* Newb. in California), account for 53% of the species identified in the selected articles. This overrepresentation may result from these species' ecological importance and economic significance in these regions. These four species often occur in open oak woodlands in these regions (Moreno et al., 2013; Roig et al., 2013). Such open oak woodlands are socio-ecological systems commonly managed as silvo-pastoral systems with high environmental and socio-economic values, which may explain the higher number of studies found on these particular oak species. In the Mediterranean Basin, *Quercus suber* L. and *Quercus ilex* L. are predominant in the Iberian landscapes, mainly in the silvopastoral systems of "montado" in Portugal, and "dehesa" in Spain. Montados and dehesas can host high biodiversity (Bugalho et al., 2011; Pinto-Correia et al., 2011), and are usually characterized by different uses such as cork harvesting, livestock production, hunting, and agriculture (Bugalho et al., 2009). *Quercus ilex* L. and *Quercus rotundifolia* Lam. in particular provide high quality acorns as food for livestock and game species, while *Quercus suber* L. generates cork, harvested periodically from the trees. Cork is a valuable sustainable material mostly used for wine bottle stoppers, but also with many other applications, which generates a solid industrial activity, being Portugal world leader in cork production (46%), followed by Spain (33%), Morocco (6%), Algeria (5%), Tunisia (4%), Italy (3%) and France (3%) (APCOR, 2020). In California, the two most addressed species (*Quercus agrifolia* Née and *Quercus kelloggii* Newb.) share broadly overlapping distributions (Hauser et al., 2017). Similarly to their Mediterranean Basin counterparts, oak woodlands in California where these species are present, serve several important ecological functions, providing important wildlife habitats and high levels of biodiversity, surpassing that of any other ecosystem in this region (Bernhardt and Swiecki, 2001). These oak woodlands are also characterized by extensive management practices that contribute to high environmental protection (Alagona et al., 2013).

We found that rare or endemic oak species were less frequently studied, potentially leading to gaps in their conservation needs. Many of these species are in critical situations or lack sufficient data for proper assessment (Carrero et al., 2020). In the Mediterranean Basin, examples of understudied species include *Quercus afares* Pomel, classified as vulnerable; *Quercus kotschyana* O.Schwarz, an endangered species found only in Lebanon, *Quercus castaneifolia* C.A.Mey, a near threatened species; and *Quercus canariensis* Willd., *Quercus hartwissiana* Steven and *Quercus ungeri* Kotschy, which have limited distributions and insufficient data to assess their present situation. Among these species, only *Quercus canariensis* Willd. was mentioned in more than one study, appearing in six articles, while the others were referenced in just one study each.

In California, we found only one article referring to *Quercus engelmanni* Greene, an endangered oak species, and six studies related to *Quercus parvula* Greene and *Quercus lobata* Née both classified as near threatened. Future research should consider increasing the focus on these species, especially those classified as endangered or with restricted distributions.

4.3. Regional variations in oak decline causal factors and biotic invasion risks

The results of this review reveal similarities and differences between the Mediterranean Basin and California in oak decline research. Further research is needed to determine if these differences correspond to actual differences in oak decline dynamics between the two regions. Pathogens were the most frequently studied causal factor in both regions, with identical proportions (Mediterranean Basin 48%, California 47%). Similarly, climate related factors received comparable research attention (Mediterranean Basin 18%, California 14%). However, California has a proportionally higher number of studied causal factors related to

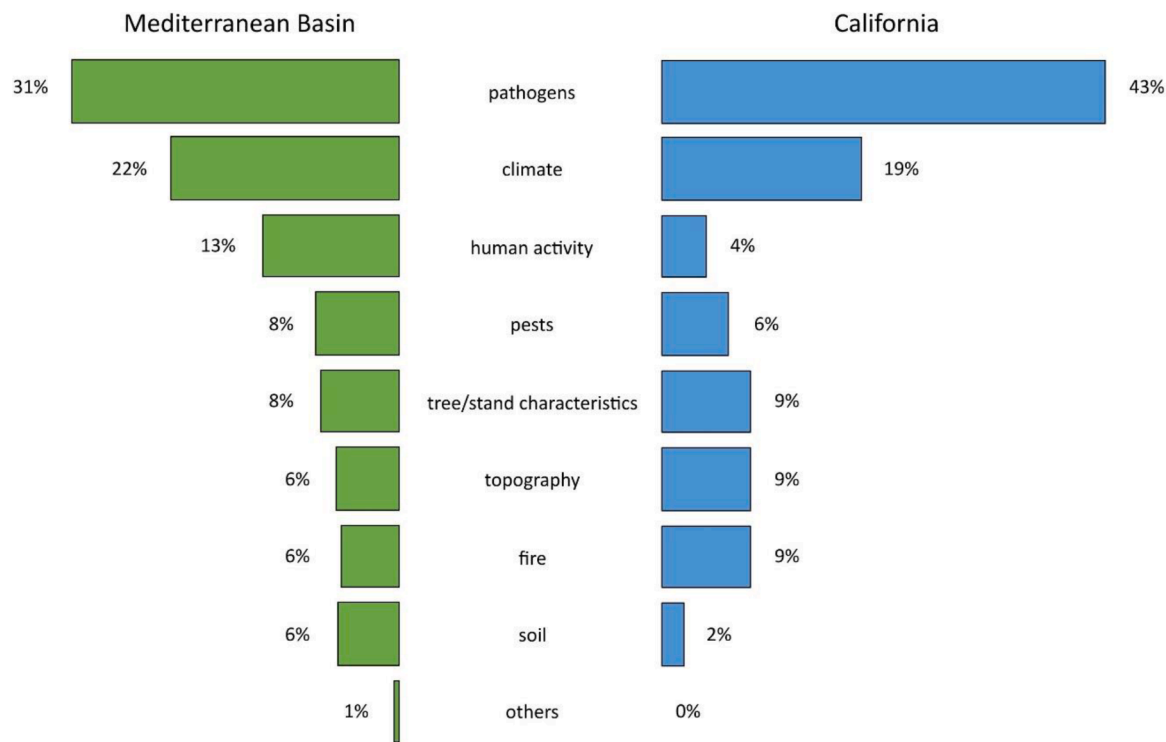


Fig. 5. Categories of causal factors of oak decline present in the select studies.

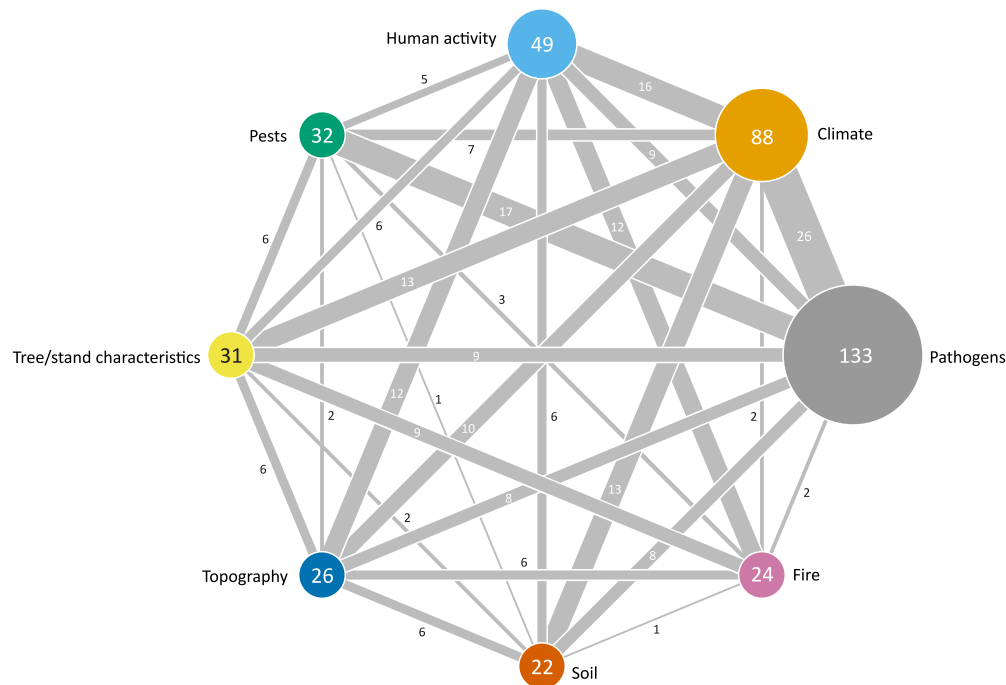


Fig. 6. Network map of co-occurrence of causal factor categories identified in the selected papers. The numbers within the nodes indicate the total number of articles that addressed each category. The lines connecting the nodes represent the overlap in research, showing the number of studies that identify the two different categories connected.

pests (Mediterranean Basin 10%, California 21%), while the Mediterranean Basin factors related to management practices were more frequently studied (Mediterranean Basin 8%, California 2%).

We also found a limited overlap in the studied fungi species between regions, with only six species common to both regions, out of a total of 108 species identified. This result suggests a strong regional specificity

in fungal pathogens associated with oak decline in each of the regions. Regarding in particular the Oomycota species, we observed a dichotomy between the two regions, with *Phytophthora cinnamomi* present in the Mediterranean Basin and *Phytophthora ramorum* in California. These regional differences in pathogen distribution highlight potential risks of biotic invasions across the Mediterranean Basin and California. Similar

climatic conditions and the presence of suitable hosts may allow biotic agents from one region to thrive and cause damage if they accidentally spread to another, which already occurred in the past. For instance, *Phytophthora ramorum*, responsible for widespread oak mortality in California, is itself of exotic origin (Rizzo and Garbelotto, 2003). Similarly, *Phytophthora cinnamomi*, a major pathogen in the Mediterranean Basin, originated in the New Guinea-Oceania region (Arentz, 2017). MTC regions are particularly susceptible to exotic forest diseases due to their richness in plant hosts, mild and wet winters that tend to favor pathogen development, dense human population, and tourism (Garbelotto and Pautasso, 2012). These conditions facilitate the introduction and spread of invasive pathogens, often with unexpected consequences (Garbelotto and Pautasso, 2012; Ghelardini et al., 2017; Liebhold et al., 1995). For instance, *Phytophthora ramorum* can successfully colonize the bark of European oaks (Moralejo et al., 2009).

Similarly, our analysis revealed strong regional variations in the communities of insect pests, with unique species assemblages in the Mediterranean Basin and unique assemblages in California. Notably, the *Cerambycidae* family was exclusively identified in Mediterranean Basin, and there were no shared species of the *Buprestidae* family between the two regions. Similarly to pathogens, the observed regional differences in pest distribution highlight the potential risk of future cross-regional invasions between the Mediterranean Basin and California, as has already been documented (Rabaglia et al., 2020; Seybold et al., 2016), which could significantly impact oak populations.

4.4. Pathogen-driven oak decline

Pathogens are the most researched causal factors of oak decline in MTC regions. Our analysis revealed, in the first place, the importance of different fungal species, particularly from the *Botryosphaeriaceae* family, in both the Mediterranean Basin and California. These pathogens cause cankers and long vascular necroses, as observed in studies from the Mediterranean Basin (Luque et al., 2000; Luque and Girbal, 1989; Sánchez et al., 2003) and California (Hecht-Poinar, 1981; Lynch et al., 2013), suggesting their association with oak decline.

Regarding Oomycota, the role of both *Phytophthora cinnamomi* and *Phytophthora ramorum* in oak decline has been extensively studied. In the Mediterranean Basin, the pathogenicity and aggressive nature of *Phytophthora cinnamomi* on oaks is well documented (Gallego et al., 1999; Luque et al., 2000, 1999; Robin et al., 1998; Scanu et al., 2013). This pathogen parasitizes the living roots of woody species, causing rot of finer feeder roots (Camilo-Alves et al., 2013). Infected trees can experience a sudden death, or a progressive decline and gradual loss of foliage (Gallego et al., 1999). The interaction of this pathogen species with abiotic factors, such as drought, causes increased stress to the affected tree when the pathogen, which may weaken the tree and ultimately lead to oak decline (Brasier et al., 1993). Alternating periods of drought and wet weather may play a key role the susceptibility of the trees to the disease and in oak decline, by creating the conditions for the pathogen dispersal (Sánchez et al., 2002) together with other factors such as topography and soil characteristics, including the presence of soil water (Sánchez-Cuesta et al., 2022). Biotic factors, including management practices, such as soil disking, and the presence of host plants such as yellow lupin (*Lupinus luteus*), were found to facilitate the propagation of the disease and to increase tree mortality associated with this pathogen (Camilo-Alves et al., 2020).

In California, oak decline is primarily associated with *Phytophthora ramorum*. Unlike *Phytophthora cinnamomi*, *Phytophthora ramorum* is found in the aerial parts of the tree, especially leaves, where it produces sporangia, disseminated by water (Henricot and Prior, 2017). This pathogen causes large stem cankers, leading to foliage browning and an apparent sudden death (Rizzo and Garbelotto, 2003). Abiotic factors such as elevated temperature and high moisture, have also been related to an increased prevalence of this pathogen in California (Cushman and Meentemeyer, 2008). Similar trends of the influence of alternating

periods of drought and wet weather seen in the dispersion of *Phytophthora cinnamomi* were observed in California with *Phytophthora ramorum* (Garbelotto et al., 2017; Kozanitas et al., 2022).

Additionally, evidence found in the selected articles also suggests that human activities, such as hiking, contribute to the spread of *Phytophthora* species. For example, *Phytophthora ramorum* was more frequently found in soil along hiking trails compared to adjacent undisturbed areas (Cushman and Meentemeyer, 2008) and its prevalence was higher in parks with greater visitor traffic (Kelly and Meentemeyer, 2002). A similar trend has been observed in the Mediterranean Basin, where the highest occurrence of oak decline was found close to urban areas and artificial infrastructures (Fernández-Habas et al., 2019). Furthermore, the activity of various species of ambrosia beetles contributes to disease spread by creating additional stress on trees and acting as vectors for pathogen transmission (McPherson et al., 2005, 2001).

4.5. Climatic factors in oak decline

Climate-related factors, including drought, elevated temperatures, and extreme precipitation events play a critical role in oak decline and mortality in MTC regions. Our analysis of the selected articles shows that these climatic factors are closely associated with oak decline, representing the second most researched category in these regions.

The typical dry summer season of Mediterranean climates, coinciding with high atmospheric temperatures, is particularly significant for oak health. Extended dry periods can significantly affect oak water stress, and reduce their resilience to drought, leading to oak decline (Caldeira et al., 2015; Conte et al., 2019; Romagnoli et al., 2018; Tognetti et al., 1998; Liu et al., 2018; Ogaya and Peñuelas, 2007, 2021; Pollastrini et al., 2019). Elevated temperatures are also associated with reduced oak growth rates and increased mortality in oak forests (Ibáñez et al., 2014; Ogaya et al., 2015; Ruiz-Benito et al., 2013).

While climate factors alone can significantly impact oak health, our analysis reveals the crucial role of interactions with biotic agents in exacerbating oak decline, such as pathogens, described in the previous section, and pests. Selected papers demonstrate that elevated temperatures combined with periods of high precipitation promote pathogen dispersal and infection, notably for *Phytophthora ramorum* (Dillon and Meentemeyer, 2019; Kozanitas et al., 2022; Meentemeyer et al., 2008) and *Phytophthora cinnamomi* (Corcobado et al., 2013a; Sánchez et al., 2002). Also, sites with warmer climates are linked to a higher incidence of ambrosia beetles, which are associated with pathogens and oak decline (Bellahirech et al., 2019).

Our analysis also revealed important interactions between climate factors and land management practices that can increase oak decline. The combination of drought, higher mean temperatures, and intensive land use management has been shown to induce oak decline (Acácio et al., 2021; Amorini et al., 1996; Aubard et al., 2019; Gea-Izquierdo et al., 2021). Interestingly, oak decline was also associated with land abandonment followed by shrub encroachment, particularly if combined with drought (Acha and Newing, 2015; Gazol et al., 2021; Hernández-Lambrano et al., 2019). In the case of *Quercus suber*, cork stripping during drought conditions is known to increase tree stress and its vulnerability to lack of water, which may debilitate trees and increase susceptibility to decline (Costa-e-Silva et al., 2021).

4.6. Insect pests in oak decline

Our analysis revealed that insect pests, particularly bark and ambrosia beetles of the family *Curculionidae* (subfamilies *Scolytinae* and *Platypodinae*), play a crucial role in oak decline across MTC regions. The *Buprestidae* and *Cerambycidae* families were also frequently mentioned in the literature, although the latter were reported exclusively in the Mediterranean Basin.

Ambrosia beetles, known for their symbiotic interaction with fungi,

dig tunnels inside tree trunks which distinguishes them from bark beetles, which primarily feed on inner bark (Catry et al., 2017). Some of the fungi species carried by ambrosia beetles are pathogenic to oaks, contributing to their decline (Inácio et al., 2022, 2021; Latifa, 2011; Lynch et al., 2014; McPherson et al., 2013). Increased oak mortality associated with these insects has been reported both in California (McPherson et al., 2005) and the Mediterranean Basin (Francardi et al., 2017). Following wildfires in the Mediterranean Basin, populations of ambrosia beetles and their attacks on mature cork oaks significantly increased (Catry et al., 2022, 2017). Moreover, bacteria associated with the ambrosia beetle *Platypus cylindrus* have been shown to contribute to oak decline in this region (Nones et al., 2022).

In California, mortality attributed to species from *Buprestidae*, such as the gold spotted oak borer (*Agrilus auroguttatus*), has been reported (Coleman et al., 2012, 2011; Coleman and Seybold, 2008), although this may also be associated with other factors such as climate and pathogens (Lynch et al., 2014). *Buprestidae* beetle larvae feed on the surface of sapwood, creating extensive galleries with a general vertical orientation (Coleman and Seybold, 2008). In the Mediterranean Basin, *Coraebus florentinus*, another *Buprestidae* species, has been associated with diverse fungal communities found in the exoskeleton of adults and larvae, potentially contributing to oak decline (Pinna et al., 2019; Torres-Vila et al., 2019).

Within the *Cerambycidae* family, the large wood borer *Cerambyx welensii* has been identified as a major cause of oak woodlands degradation and oak mortality in certain regions of Spain (Domínguez et al., 2022). Additionally, research shows that the dispersal of *Botryosphaera* pathogens, which are related to oak decline, is facilitated by insect vectors such as *Cerambyx welensii*, *Coraebus fasciatus* (Panzavolta et al., 2017), *Coraebus florentinus* (Pinna et al., 2019) and *Platypus cylindrus* (Latifa Belhoucine, 2011) in the Mediterranean Basin.

4.7. Management and oak decline

Human activities, particularly forest management practices, were identified in this review as the fourth most frequently researched causal factor of oak decline. Both Mediterranean Basin and California landscapes, and specifically their oak forests, have been shaped by anthropogenic activities throughout history, dating back thousands of years in the Mediterranean Basin (Gil-Pelegrín et al., 2017a). Although we did not find hints in the selected literature that connect this historical long-term use to present vulnerability, in fact, in recent decades, human pressures such as land-use changes, intensive management practices and grazing have had a significant impact on oak forest ecosystems.

Among management practices, cattle overgrazing is often highlighted as a significant cause, impacting adult trees in oak silvopastoral systems (Bugalho et al 2009; Alaoui et al., 2020; Corcobado et al., 2015; Laouina et al., 2010; Pinto-Correia and Azeda, 2017; Pinto-Correia and Mascarenhas, 1999; Wojterski, 1990; Roula et al., 2020). High stocking rates can induce soil compaction, damaging soil structure and tree root systems (Pinto-Correia and Mascarenhas, 1999).

Other management practices such as shrub clearing using heavy machinery, can also lead to soil compaction (Corcobado et al., 2015) reduce soil aeration, water infiltration, and reduce tree vitality (Camilo-Alves et al., 2020). In Northern African Mediterranean countries, the overuse of wood for fuel is considered a primary contributor to oak area degradation (Alaoui et al., 2020; Laouina et al., 2010; Wojterski, 1990). For *Quercus suber* L., cork harvesting may negatively affect growth metabolism, particularly in vulnerable trees (Costa-e-Silva et al., 2021). In addition, improperly conducted cork harvesting may also play a role in the infestation of trees by xylophagous insects, as their presence is strongly correlated with bark removal and tree wounds (Catry et al. 2022; Younsi et al., 2021).

Both over-use and lack of management may negatively affect the sustainability of oak ecosystems (Bugalho et al 2011; Acha and Newing, 2015; Pinto-Correia and Mascarenhas, 1999). Shrub encroachment

resulting from management neglect has been associated with higher oak mortality rates in multiple studies (Acácio et al., 2009; Costa et al., 2009; Hernández-Lambrano et al., 2019; Sucena-Paiva et al., 2022).

Land-use change also plays a critical role in the decline of oak cover. In Portugal, long-term analyses reveal that various types of oak forests have been gradually replaced by other forest species, agricultural lands and shrublands (Acácio et al., 2021, 2017). In Northern Africa (Algeria), land-use changes, grazing and recurrent fires have also contributed to the disappearance of oak forests (Bardadi et al., 2021). Research in Spain has shown that land-use changes created isolated forest patches, which combined with poor silvicultural practices may have increased the occurrence of oak pests and created the conditions to more rapid deterioration of oak ecosystems (Domínguez et al., 2022). California's oak woodlands have also undergone a decline in extent and density, driven by the establishment of fruit and nut orchards in the late 19th century and the rapid urban expansion following World War II, (Whipple et al., 2011).

Finally, our analysis revealed a notable gap in research concerning the effects of pollution on oak decline. Of the limited studies available, one reported oak dieback near an industrialized area in Spain, where high tree mortality was observed (Morillas et al., 2012) and another suggested the influence of pollutants on oak decline in coastal areas of Italy (Calistri et al., 1993). Additionally, ozone exposure has been shown to stimulate fungal leaf attacks, which may further contribute to oak decline (Paoletti et al., 2007).

4.8. Topography and oak decline

Topographic and soil characteristics can play an important role in oak decline, with their complex interactions influencing tree health through various mechanisms related to water availability, pathogen activity, and management practices. Increased oak decline has been documented in lowland areas with flat or gentle slopes, which are prone to water accumulation, fostering moisture conditions favorable to higher pathogen activity (Dillon and Meentemeyer, 2019; Fernández-Habas et al., 2019; Huesca et al., 2021; Moreira and Martins, 2005). Furthermore, gentle slopes are often associated with intensified land management, which may damage oak roots and increase tree susceptibility (Acácio et al., 2021; Herguido Sevillano et al., 2017).

Additionally, steeper slopes can accelerate soil erosion due to lower water retention and thinner soils, both of which can limit tree vitality and contribute to oak decline (Camilo-Alves et al., 2020; Moreira et al., 2007).

The interaction between slope and aspect and their combined influence on oak decline is also addressed in the literature (Camilo-Alves et al., 2020; Costa et al., 2010; Hernández-Lambrano et al., 2019; Moreira and Martins, 2005). Higher oak mortality is typically associated with south and southwest-facing slopes (Calistri et al., 1993; Camilo-Alves et al., 2020; Costa et al., 2010; Hernández-Lambrano et al., 2019; Huesca et al., 2021; Moreira and Martins, 2005; Peñuelas et al., 2000; Torres-Vila et al., 2019). The increased sun exposure on south-facing slopes results in higher temperatures, elevated evaporation, and greater water loss, thereby exacerbating drought conditions and oak decline (Peñuelas et al., 2000).

Interestingly, one study observed oak decline on northern exposures, combined with lower slopes and soil humidity, which was related to the presence of the pathogen *Phytophthora cinnamomi* (Sánchez-Cuesta et al., 2022). A similar case was identified in California (Kozanitis et al., 2022), where east-facing slopes functioned as reservoirs for the pathogen *Phytophthora ramorum*.

Shallow soils, particularly on steep slopes, have also been associated with oak decline in Portugal (Aubard et al., 2019; Moreira and Martins, 2005), Spain (Doña et al., 2022; Peguero-Pina et al., 2015) and Türkiye, in association with a high water deficit (Kumbasli et al., 2017). Conversely, soil properties and topographic factors that increase soil moisture, such as rich soils and wet conditions, may also exacerbate

infection by *Phytophthora cinnamomi* and contribute to oak decline (Corcobado et al., 2013b; Sánchez-Cuesta et al., 2022, 2021).

4.9. Fire and oak decline

Fire is a major disturbance in MTC regions, where fire regimes have been changing in recent decades toward more frequent and severe events (Pausas and Fernández-Muñoz, 2012; Ruffault et al., 2018). Despite this, fire was not identified as a primary causal factor in oak decline within the selected studies, probably because many Mediterranean oak species can withstand or resprout after fire (Catry et al., 2012; Pausas, 2015). However, studies show that fire can directly cause oak mortality (Catry et al., 2013, 2010), or it can predispose trees to subsequent attacks by pathogens or pests, thereby increasing overall decline (Catry et al., 2022, 2009).

The recurrence of fires can trigger ecological changes, fostering more fire-prone landscapes with pyrophytic shrub encroachment, and contributing to the degradation of oak woodlands (Acácio et al., 2021, 2010, 2009; Godinho et al., 2016). In the Mediterranean Basin, research on *Quercus suber* has linked management practices to post-fire oak mortality, showing that bark thickness and cork harvesting significantly influence the species' resistance to fire (Catry et al., 2009, 2012, 2013; Moreira et al., 2007, 2009).

In California, where research on fire as a cause of oak mortality is more limited, fire is recognized as a recurrent threat to hardwoods, with variable impacts on different oak species (Long et al., 2018). In this region, mortality rates following fire have been shown to increase with drought exposure (Barker et al., 2022), and fire may contribute to the containment of *Phytophthora ramorum* during post-fire forest recovery (He et al., 2021).

Fire-induced oak mortality also appears to be influenced by topographic factors such as aspect. For instance, higher oak mortality or lower crown regeneration following fire have been documented on southern exposures, which correspond to drier locations and may cause higher physiological stress (Catry et al., 2009, 2012; Huesca et al., 2021; Moreira et al., 2007). However, Barker et al. (2022) found similar mortality risk on both southwestern and northeastern exposures, with decreasing mortality as slope increased, underscoring the complexity of fire effects.

5. Conclusions

This systematic review demonstrates the complex and multifaceted nature of oak decline in MTC regions, highlighting the critical interplay between biotic and abiotic factors. Pathogens and climatic factors emerged as the factors that interact most frequently, while pathogens were also the most frequently studied in isolation. Our comprehensive analysis of 241 articles published between 1981 and 2022 reveals significant temporal and geographical variations in oak decline research, emphasizing the need for a holistic approach to understand and mitigate this problem.

Future research should emphasize the interaction of distinct factors and regional comparisons, with higher knowledge interchange among Mediterranean regions, including Northern African countries, Europe and California. Furthermore, the integration of scientific knowledge with stakeholders' perspectives and traditional practices is essential for developing effective strategies to mitigate oak decline in the Mediterranean regions, where ecological and human-use similarities are prevalent. Finally, expanding research should cover a broader range of *Quercus* species.

In conclusion, our systematic review underscores the need for a comprehensive, interdisciplinary approach to research and management. By addressing the identified research gaps and fostering greater collaboration across regions, we can enhance our understanding of oak decline and develop more effective strategies for its mitigation. The implications of this review may well be extended beyond these regions,

advancing in the understanding of oak decline, and contributing to the preservation and sustaining of oak forests in other regions.

CRedit authorship contribution statement

Marco Marques: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Miguel N. Bugalho:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Vanda Acácio:** Writing – review & editing, Supervision, Methodology. **Filipe X. Catry:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.tfp.2025.100803.

Data availability

I have shared my data as a Supplementary File (File 1)

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